

IoT Course — Hands-On #7

ThingsBoard Hello World — Assignment #4

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Platform: thingsboard.cloud | Date: 2026-05-29

1. Overview

This report documents the step-by-step completion of the ThingsBoard Hello World assignment. ThingsBoard is a cloud-based IoT platform that provides device management, real-time data visualization, alarm management, and multi-tenant access control. The assignment walks through provisioning a device, sending telemetry, building a dashboard, setting up alarms, and creating a customer user with restricted access.

Platform: ThingsBoard Cloud (thingsboard.cloud)

Authentication: GitHub OAuth

Device: Thermometer (Device profile: default)

Protocol: HTTP POST via cURL (Check connectivity tool)

2. Login to ThingsBoard Cloud

Signed up at thingsboard.cloud and logged in via GitHub OAuth. The ThingsBoard Tenant interface homepage confirms successful authentication, showing the Alarms and Devices summary panels.

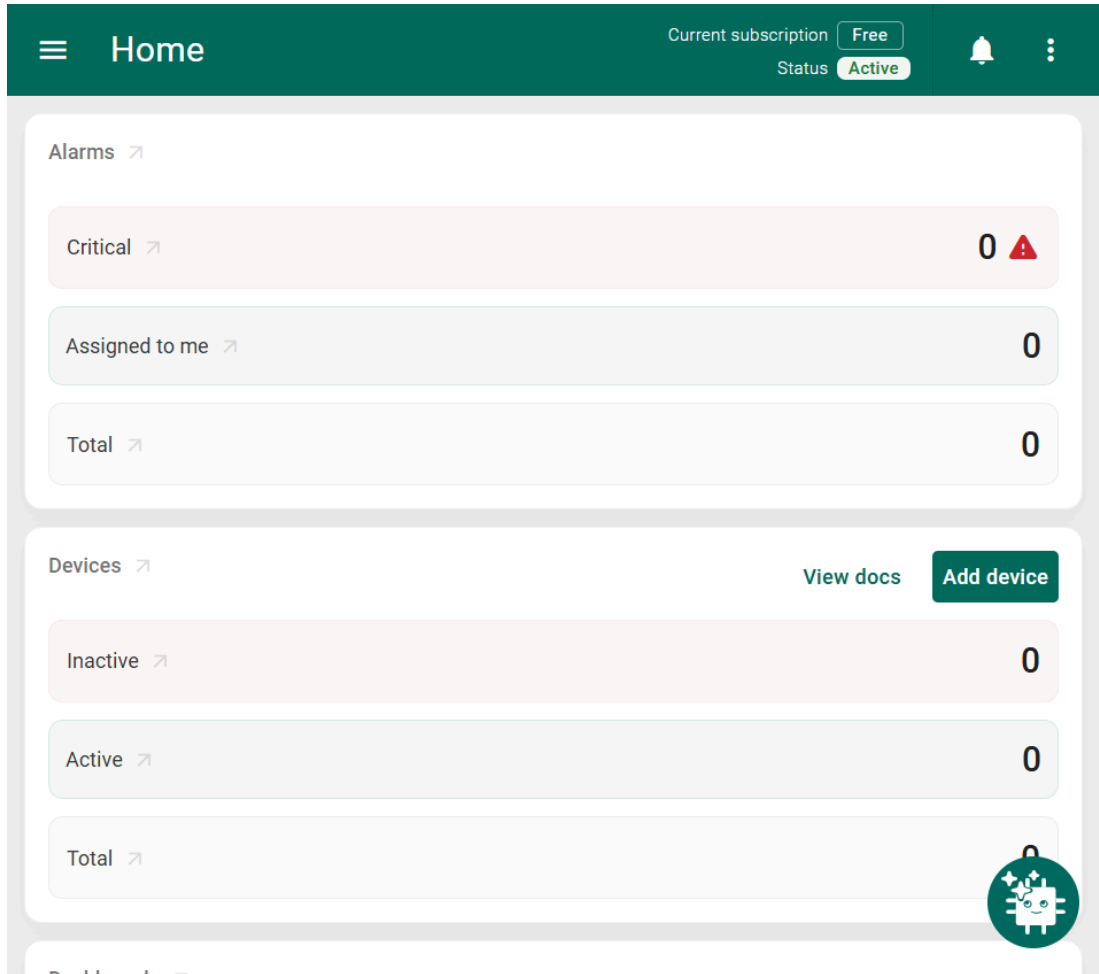


Figure 1: ThingsBoard homepage after login — tenant dashboard with Alarms and Devices panels

3. Step 1 — Device Provisioning & Telemetry

3.1 Device Created

Created a device named "Thermometer" via Entities → Devices → Add new device. The device was configured with the default device profile. Upon creation, ThingsBoard presented the "Check connectivity" dialog to verify the connection.

Device name: Thermometer

Device profile: default

Protocol used: HTTP (cURL on Windows)

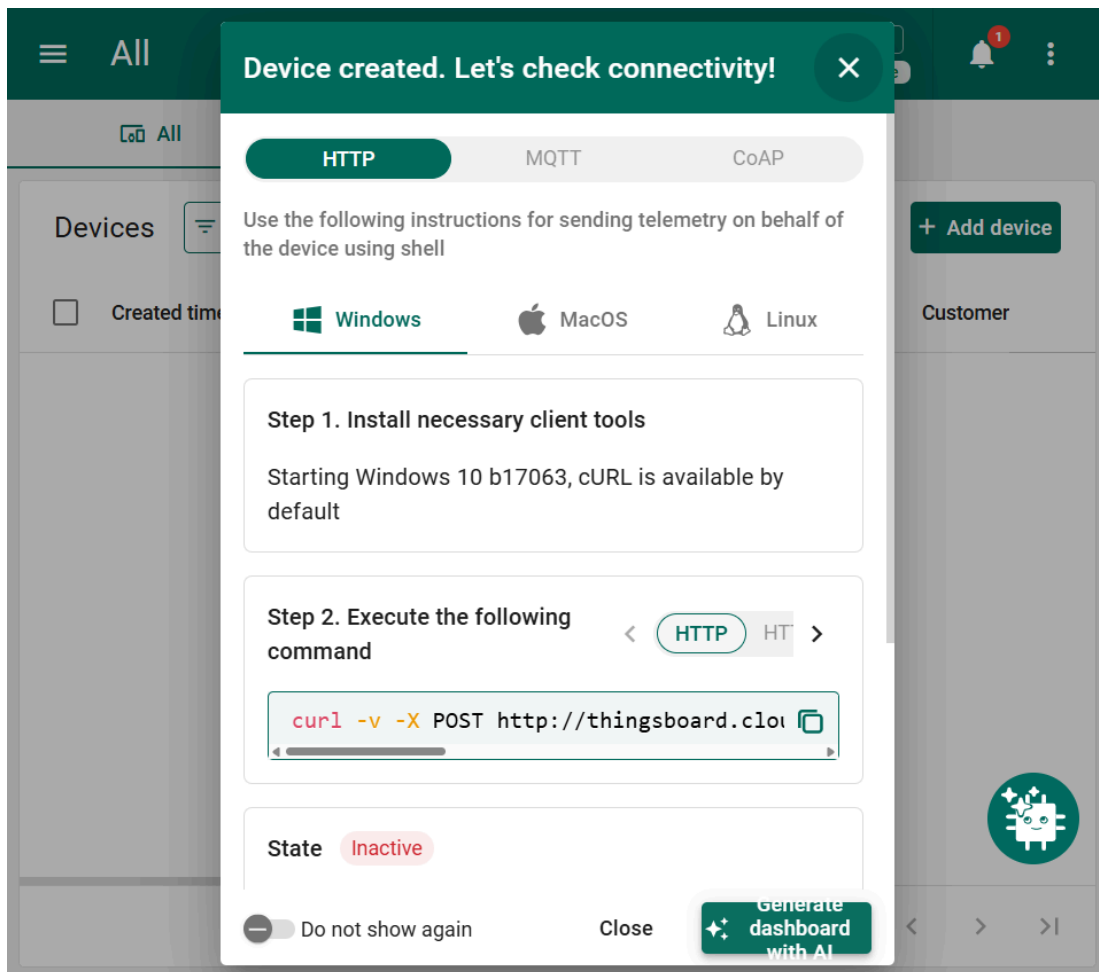


Figure 2: Check connectivity dialog — HTTP cURL command shown, device state: Inactive

3.2 Telemetry Active

Executed the cURL HTTP POST command from the Check connectivity tool to send the first telemetry reading (temperature=25). The device state immediately changed from Inactive to Active, and the Latest Telemetry panel confirmed the data was received at 2026-05-29 18:55:46.

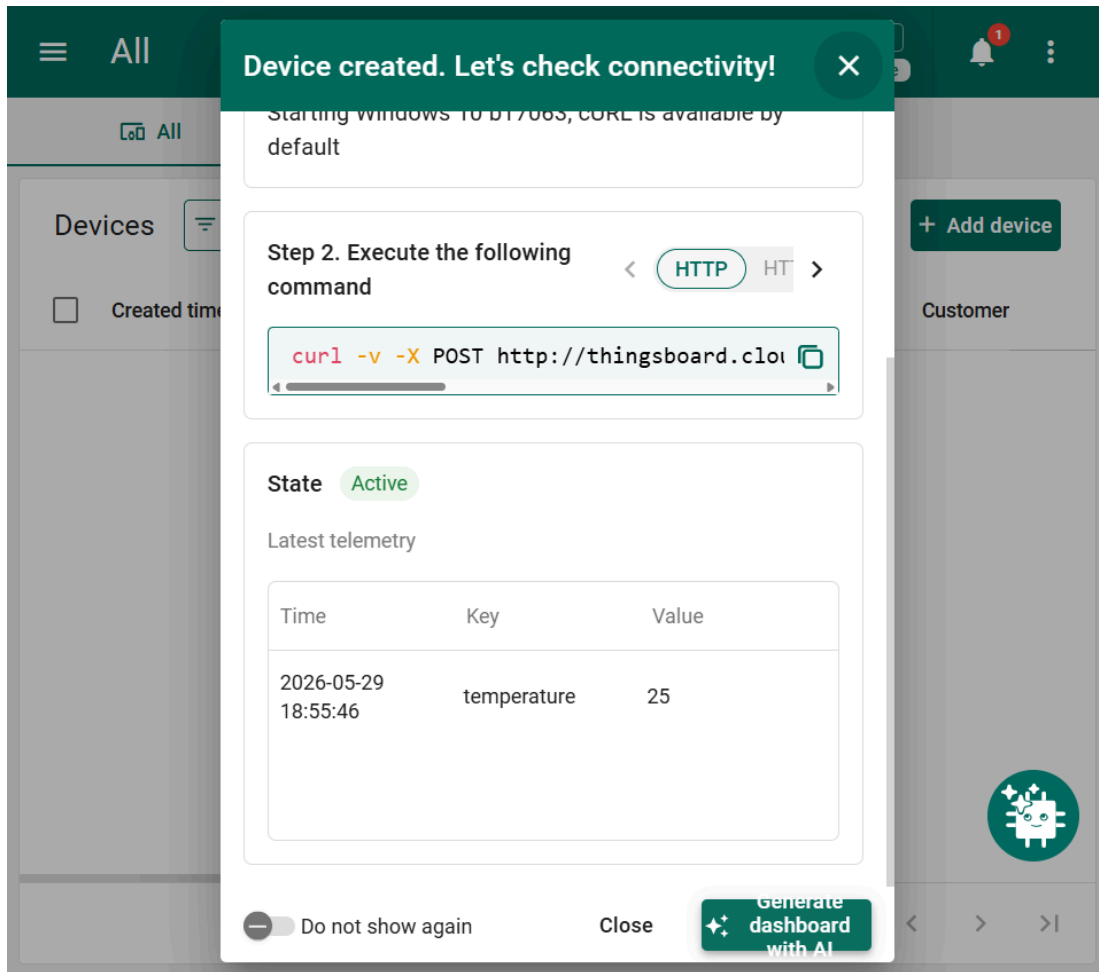


Figure 3: Device state Active — Latest Telemetry shows temperature=25 received successfully

4. Step 2 — Real-Time Dashboard

Created "My Dashboard" with three widgets displaying live data from the Thermometer device: a Value card showing current temperature (25°C), a Time series chart tracking historical temperature readings, and an Alarms table. The dashboard reflects data in real time as new telemetry arrives.

Dashboard name: My Dashboard

Widgets: Value card, Time series chart, Alarms table

Data source: Thermometer device

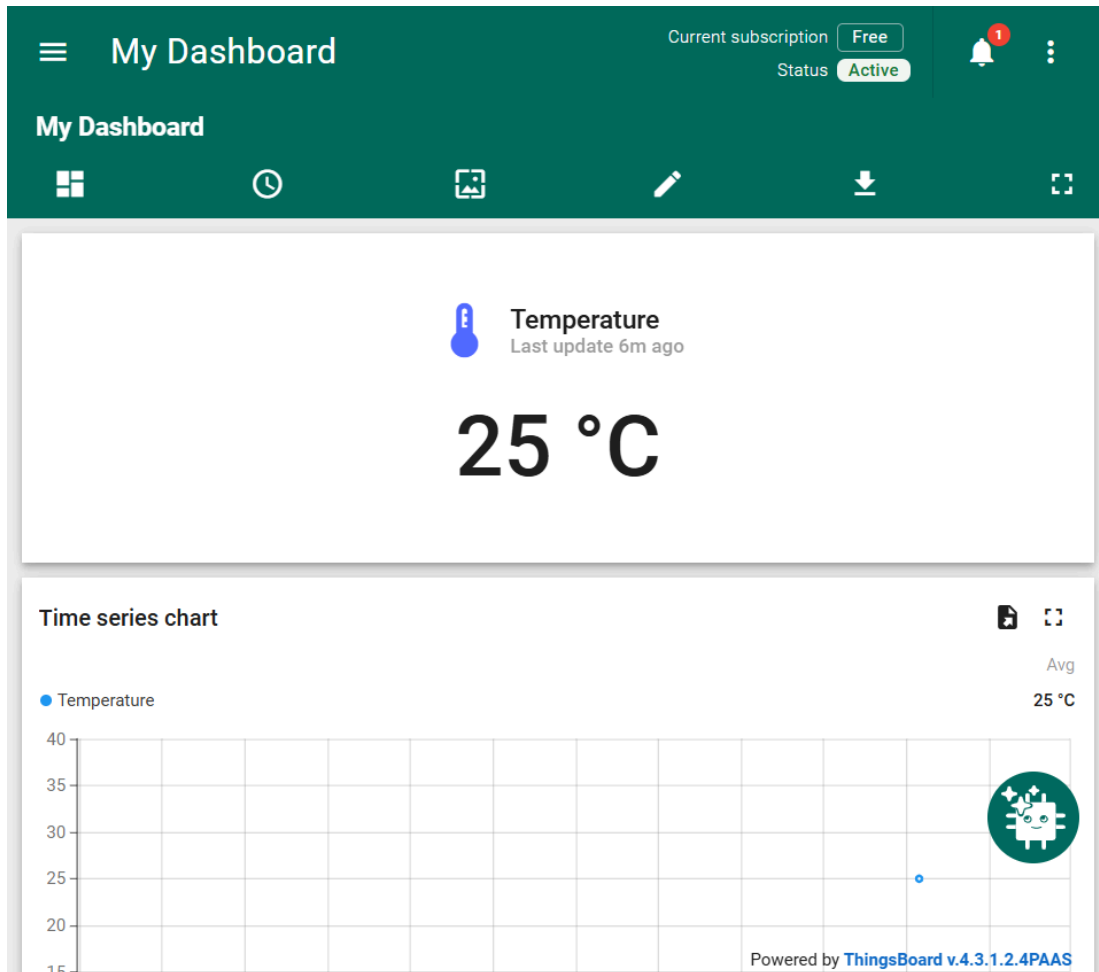


Figure 4: My Dashboard — Value card showing 25°C and Time series chart with live temperature data

5. Step 3 — Alarm Rule & Trigger

5.1 Alarm Rule Defined

Created a "High temperature" alarm rule via Alarms → Alarm rules → Add alarm rule. The condition uses Script mode with the expression: `return temperature > 25`. The rule is assigned to the default device profile with Critical severity.

Alarm type: High temperature

Condition: `return temperature > 25` (Script mode)

Severity: Critical

Entity type: Device profile: default

The screenshot shows the 'Alarm rules' management interface. At the top, there's a header with a menu icon, the title 'Alarm rules', and subscription status 'Free' and 'Active'. Below the header, there are tabs for 'Alarm list' and 'Alarm rules', with 'Alarm rules' being the active tab. The main area contains a table of alarm rules. The table has columns for 'Created time', 'Alarm type', 'Entity type', 'Entity', 'Severities', and 'Clear conditi'. A single rule is listed: 'High temperature' with 'Critical' severity and 'default' entity type. The rule was created on '2026-05-29 19:10:36'. At the bottom, there's a pagination bar showing 'Items per page: 10' and '1 - 1 of 1'.

Created time ↓	Alarm type	Entity type	Entity	Severities	Clear conditi
2026-05-29 19:10:36	High temperature	Device profile	default	Critical	☐ ⋮

Figure 5: Alarm rules list — High temperature rule configured with Critical severity on default profile

5.2 Alarm Triggered

Sent temperature=26 via the Check connectivity HTTP POST (above the threshold of 25). ThingsBoard raised a Critical "High temperature" alarm, visible in the Alarm list. The originator is the Thermometer device and the alarm was created at 2026-05-29 19:16:13.

Alarm list

Current subscription

Free

Status

Active

2

Alarm list

Alarm rules

Filter: Active

For all time

☐	Created time ↓	Originator	Type	Severity	Assignee	Details
☐	2026-05-29 19:16:13	Thermometer	High temperature	Critical	Unassigned	▼ ...

Items per page: 10

1 - 1 of 1

Figure 6: Alarm list — Critical "High temperature" alarm triggered by Thermometer at temperature=26

6. Step 4 — Customer Created

Created customer "My New Customer" via Customers → Add customer. The Thermometer device and My Dashboard were assigned to this customer using the "Manage owner and groups" option on each entity, making them accessible to customer-level users.

Customer name: My New Customer

Assigned entities: Thermometer device, My Dashboard

The screenshot displays a web application interface for managing customers. At the top, a dark green header bar contains a menu icon, the text 'All', and subscription details: 'Current subscription Free' and 'Status Active'. A notification bell icon with a red '3' is also present. Below the header, a light gray bar offers navigation options: 'All' (selected), 'Groups', and 'Hierarchy'. The main content area is titled 'Customers' and includes a toggle switch for 'Include customer entities' (which is turned on), a refresh icon, a search icon, and a '+ Add customer' button. A table lists the customers with columns for 'Created time', 'Title', 'Email', 'Country', and 'City'. One customer is listed: 'My New Customer' created on '2026-05-29 19:17:29'. A green circular icon with a robot-like face is in the bottom right corner of the table area. At the bottom, a pagination bar shows 'Items per page: 10', '1 - 1 of 1', and navigation arrows.

☐	Created time ↓	Title	Email	Country	City
☐	2026-05-29 19:17:29	My New Customer			

Figure 7: Customers list — "My New Customer" created and visible in the tenant customer list

7. Step 5 — Customer User Activated

Created a customer user account (customer.test.iot@yopmail.com) under "My New Customer" and activated it via the "Display activation link" method. After activation, the customer user login shows a restricted interface — only Home, Trendz analytics, Resources, and Notification center are available. All admin features (Devices, Dashboards, Alarms management, Customers) are hidden, confirming the correct limited customer role.

User email: customer.test.iot@yopmail.com

Activation method: Display activation link

Visible menu items: Home, Trendz analytics, Resources, Notification center

Hidden items: All admin and tenant management features

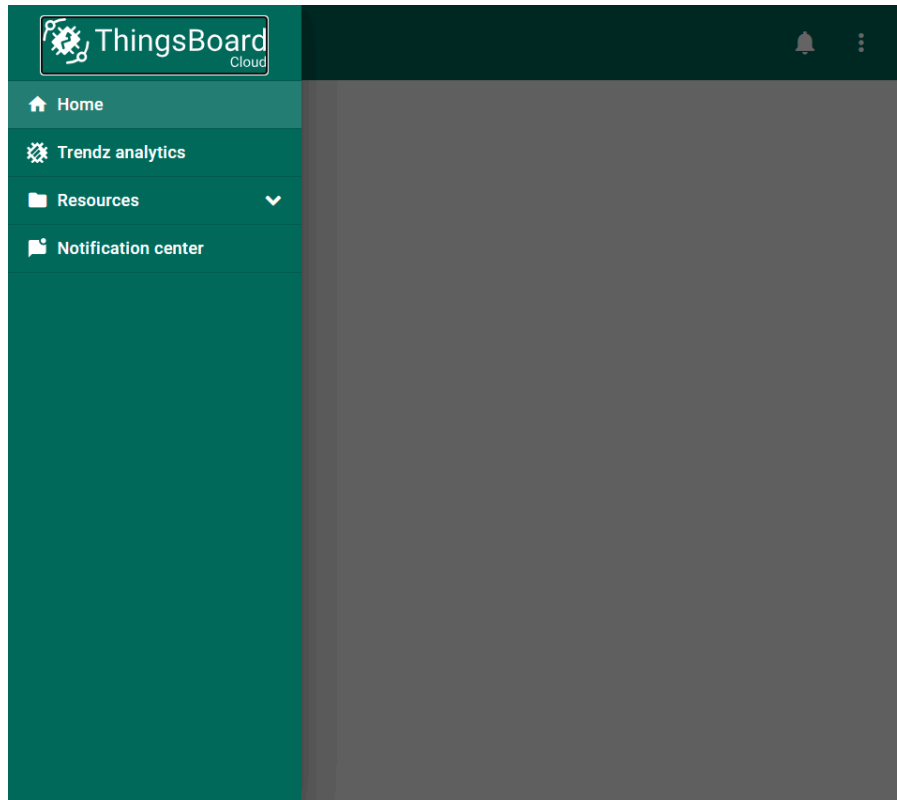


Figure 8: Customer user view — limited menu confirms restricted role (no admin features)

8. Summary

All five steps of the ThingsBoard Hello World assignment were completed successfully on the ThingsBoard Cloud platform:

- Logged in via GitHub OAuth and accessed the Tenant dashboard
- Provisioned the "Thermometer" device and sent temperature=25 via HTTP POST
- Built "My Dashboard" with Value card, Time series chart, and Alarms table widgets
- Created a "High temperature" alarm rule (threshold > 25, Critical severity) and triggered it with temperature=26
- Created customer "My New Customer", assigned the device and dashboard, and activated a customer user with restricted access

The assignment demonstrates the full ThingsBoard workflow: device provisioning → telemetry ingestion → visualization → alarm management → multi-tenant access control.